Chaoyi **Pan**

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Education

Carnegie Mellon University (CMU) — *Pittsburgh, United States*
Ph.D. Candidate in Electronic and Computer Engineering — *Aug. 2023 - now*

Tsinghua University (THU) — *Beijing, China*
Bachelor of Electronic Engineering — *Sep. 2019 - July 2023*

Research Interests

My research interests lie at the intersection of **physics-based robot data generation** and the **understanding of generative models in control**.

Honors & Awards

2025	**ICRA 2025 Best Paper Finalist**, IEEE	*Atlanta, GA*
2025-2026	**William J. Happel Fellowship**, Department of Electrical and Computer Engineering, Carnegie Mellon University	*Pittsburgh, PA*
2024-2025	**Hsu Chang Memorial Fellowship**, Carnegie Mellon University	*Pittsburgh, PA*
2021-2022	**National Scholarship (0.2%)**, Tsinghua University	*China*
2020-2021	**National Scholarship (0.2%)**, Tsinghua University	*China*
2019-2020	**National Scholarship (0.2%)**, Tsinghua University	*China*
2020	**First Prize (1%)**, Chinese Undergraduate Students' Physics Competition	*China*
2020	**First Prize (1%)**, Hardware Design Competition	*China*
2021	**Spark Innovative Talent Cultivation Program Member (2/256)**, Tsinghua University	*China*
2022	**Undergraduate Visiting Research Program Member (2/256)**, Stanford University	*Stanford*
2022	**Tsinghua University Future Scholar (1/256)**, Tsinghua University	*China*

Publications (* equal contribution, † equal advising)

Journal Articles

Global Convergence of Sampling-Based Nonconvex Optimization through Diffusion-Style Smoothing
Zeji Yi*, **Chaoyi Pan***, Guanya Shi, Guannan Qu
Transactions on Machine Learning Research (Accepted) (2026)

Conference Proceedings

Learning Gentle Humanoid Locomotion and End-Effector Stabilization Control
Yunfei Li, Yuanhang Zhang, Wenli Xiao, **Chaoyi Pan**, Huazhe Weng, Guanqi He, Tairan He, Guanya Shi
Conference of Robot Learning, 2026

Much Ado About Noising: Dispelling the Myths of Generative Robotic Control
Chaoyi Pan, Giri Anantharaman, Nai-Chieh Huang, Claire Jin, Daniel Pfrommer, Chenyang Yuan, Frank Permenter, Guannan Qu, Nicholas Boffi, Guanya Shi, Max Simchowitz
International Conference on Learning Representations, 2026

SPIDER: Scalable Physics-Informed Dexterous Retargeting
Chaoyi Pan, Changhao Wang, Haozhi Qi, Zixi Liu, Homanga Bharadhwaj, Akash Sharma, Tingfan Wu, Guanya Shi†, Jitendra Malik, Francois Hogan
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026

Whole-Body Model-Predictive Control of Legged Robots with MuJoCo
John Z Zhang, Taylor A Howell, Zeji Yi, **Chaoyi Pan**, Guanya Shi, Guannan Qu, Tom Erez, Yuval Tassa, Zachary Manchester
IEEE International Conference on Robotics and Automation, 2026

ASAP: Aligning Simulation and Real-World Physics for Learning Agile Humanoid Whole-Body Skills
Tairan He, Jiawei Gao, Wenli Xiao, Yuanhang Zhang, Zi Wang, Jiashun Wang, Zhengyi Luo, Guanqi He, Nikhil Sobanbabu, **Chaoyi Pan**, Zeji Yi, Guannan Qu, Kris Kitani, Jessica Hodgins, Linxi "Jim" Fan, Yuke Zhu, Changliu Liu, Guanya Shi
Conference of Robot Learning, 2025

Full-Order Sampling-Based MPC for Torque-Level Locomotion Control via Diffusion-Style Annealing
Haoru Xue*, **Chaoyi Pan***, Zeji Yi, Guannan Qu, Guanya Shi
IEEE International Conference on Robotics and Automation, ***Best Paper Final List***, 2025

Model-based Diffusion for Trajectory Optimization
Chaoyi Pan*, Zeji Yi*, Guanya Shi†, Guannan Qu†
Advances in Neural Information Processing Systems, 2024

CoVO-MPC: Theoretical Analysis of Sampling-based MPC and Optimal Covariance Design
Zeji Yi*, **Chaoyi Pan***, Guanqi He, Guannan Qu†, Guanya Shi†
Learning for Dynamics and Control Conference, 2024

Efficient Bimanual Handover and Rearrangement via Symmetry-Aware Actor-Critic Learning
Yunfei Li*, **Chaoyi Pan***, Huazhe Xu, Xiaolong Wang, Yi Wu
IEEE International Conference on Robotics and Automation, 2023

In-Hand Manipulation of Unknown Shapes with Tactile Sensing
Chaoyi Pan*, Marion Lepert*, Shenli Yuan, Rika Antonova, Jeannette Bohg
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2023

Flexible Decentralized Displacement-Based Formation Control: An Multi-agent Reinforcement Learning Approach
Chaoyi Pan, Yuzi Yan, Zexu Zhang, Yuan Shen
European Signal Processing Conference, 2022

Research Experience

Amazon Frontier AI and Robotics Lab — *San Francisco, USA*
Research Intern — *Jan 2026 - Present*
• Working on understanding of action representation and large-scale behavior cloning data collection under supervision of Guanya Shi and Rocky Duan

Meta FAIR Embodiment & Actions Research Team — *Pittsburgh, USA*
Research Science Intern — *June 2025 - August 2025*
• Developed **universal human to robot physics-based retargeting system** to convert raw human object interaction trajectory to feasible robot trajectory under supervision of Francois Hogan

Carnegie Mellon University — *Pittsburgh, USA*
Ph.D. Student — *June 2023 - Present*
• Developed **generative model for control**, bridging learning-based generative models with model-based control for contact-rich real-world tasks under supervision of Prof. Guanya Shi and Prof. Guannan Qu

Stanford Interactive Perception and Robot Learning Lab — *Stanford, USA*
Summer Intern — *June 2022 - September 2022*
• Implemented **tactile-based in-hand manipulation system** using Bayesian optimization without object shape prior or vision information under supervision of Prof. Jeannette Bohg

Tsinghua WuLab — *Beijing, China*
Research Assistant — *September 2021 - June 2022*
• Developed **bimanual coordination system** for handover and rearrangement tasks using structured reinforcement learning under supervision of Prof. Yi Wu

Tsinghua SLab — *Beijing, China*
Research Assistant — *October 2020 - September 2021*
• Designed **decentralized formation control system** for mobile robots using MARL and Hausdorff distance under supervision of Prof. Yuan Shen

Academic Service

Reviewer AAAI 2024, CoRL 2024, ICLR 2024, ICLR 2025, ACC 2025, CoRL2025, NeurIPS 2025, ICRA 2026, ICLR 2026

Skills

Programming	Python, Jax, PyTorch, Matlab, C/C++
Robotics Tools	Mujoco, IsaacGym, PyBullet, ROS, ROS2, Gazebo
Hardware Platforms	Franka Emika Panda, Roller Grasper Dexterous Hand, Unitree Go2, Unitree H1, Unitree G1, Fourier GR-1
Drawing and Typesetting	Photoshop, Illustrator, Office, LaTeX

Invited Talks

CMU Locomotion Seminar — *Pittsburgh, USA*
Full-Order Sampling-Based MPC for Torque-Level Locomotion Control via Diffusion-Style Annealing [Link] — *Oct. 2024*

NTU CARTIN Seminar — *Singapore*
Model-Based Diffusion for Trajectory Optimization and Model-Predictive Control — *Nov. 2024*